



FAMILY NAME and Firstname

MCQ DSIA Machine Learning 2
April 2026
Given by John Doe

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This is a multiple-choice exam with single or multiple correct answers. Multiple-choice questions are indicated by the ♣ symbol. For each multiple-choice question, there may be several, one, or no correct answers.

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CHECKED



UNCHECKED



ENCODE 1.234

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☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

Instructions: No documents allowed. The use of electronic calculators and cell phones is forbidden.

Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.

Question 2 This is a numeric question where you must calculate the sum and difference of -1.6038 and 0.0641
Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
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☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



+1/2/59+

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=2$, that is x^2 ,

- ☐ The derivative is $1x^1$
- ☐ The derivative is $\frac{1}{1}x^1$
- ☐ The derivative is $2x^1$
- ☐ The derivative is $\frac{1}{2}x^2$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	13	1
1	9	6

Give the recall (tolerance ± 0.01)

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

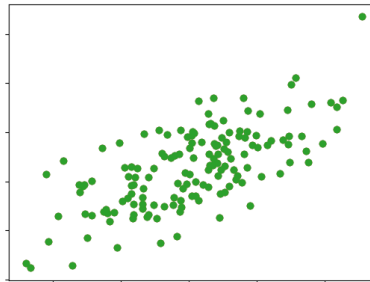


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 2.913` and `- reg.coef_ = 1.479`. For a new observation $x = 0.945$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9	
	.		0		1		2		3		4		5		6		7		8		9
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = -1.666$ and $\beta = 0.344$

- ☐ For $x=1.324$, the prediction is -1.211
- ☐ For $x=-0.512$, the prediction is 1.197
- ☐ The slope is positive



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CHECKED



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ENCODE 1.234

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☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

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Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.

Question 2 This is a numeric question where you must calculate the sum and difference of -0.8603 and 0.5195
Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
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☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



+2/2/56+

		0		1		2		3		4		5		6		7		8		9	
	.																				
		0		1		2		3		4		5		6		7		8		9	
		0		1		2		3		4		5		6		7		8		9	
+		0		1		2		3		4		5		6		7		8		9	
-		0		1		2		3		4		5		6		7		8		9	

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=3$, that is x^3 ,

- ☐ The derivative is $\frac{1}{2}x^2$
- ☐ The derivative is $2x^2$
- ☐ The derivative is $3x^2$
- ☐ The derivative is $\frac{1}{3}x^3$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	36	7
1	0	11

Give the recall (tolerance ± 0.01)

+																				
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

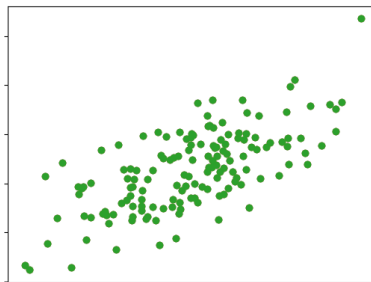


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.435` and `- reg.coef_ = 1.733`. For a new observation $x = 0.520$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

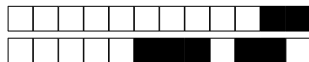
		0		1		2		3		4		5		6		7		8		9	
.			0		1		2		3		4		5		6		7		8		9
+			0		1		2		3		4		5		6		7		8		9
-			0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = -1.002$ and $\beta = 0.268$

- ☐ The slope is positive
- ☐ For $x=0.767$, the prediction is -0.501
- ☐ For $x=1.191$, the prediction is -0.683



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.	☐	0	☐	1	☐	☒	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
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☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

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- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.

Question 2 This is a numeric question where you must calculate the sum and difference of -1.1574 and 0.6963

Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	+	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



+3/2/53+

		0		1		2		3		4		5		6		7		8		9	
	.																				
		0		1		2		3		4		5		6		7		8		9	
+		0		1		2		3		4		5		6		7		8		9	
-		0		1		2		3		4		5		6		7		8		9	

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=5$, that is x^5 ,

- ☐ The derivative is $\frac{1}{5}x^5$
- ☐ The derivative is $\frac{1}{4}x^4$
- ☐ The derivative is $5x^4$
- ☐ The derivative is $4x^4$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	0	1
1	5	32

Give the recall (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

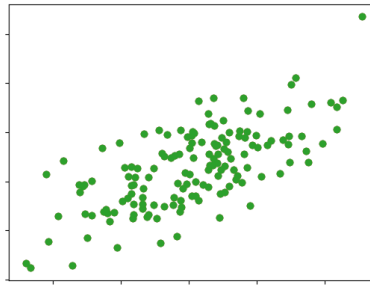


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = -0.679` and `- reg.coef_ = 0.621`. For a new observation $x = 1.331$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

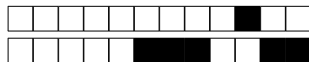
		0		1		2		3		4		5		6		7		8		9
	.																			
	+																			
	-																			

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = 0.259$ and $\beta = -0.481$

- ☐ For $x = -2.492$, the prediction is -1.126
- ☐ For $x = -0.877$, the prediction is 0.681
- ☐ The slope is positive



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ENCODE 1.234

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☐	0	☒	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
.	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐
☒	+	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8
☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8

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- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.

Question 2 This is a numeric question where you must calculate the sum and difference of -0.5055 and -1.2831

Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐
☐	+	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐
☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐

Difference:



+4/2/50+

		0		1		2		3		4		5		6		7		8		9	
	.																				
		0		1		2		3		4		5		6		7		8		9	
+		0		1		2		3		4		5		6		7		8		9	
-		0		1		2		3		4		5		6		7		8		9	

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=3$, that is x^3 ,

- ☐ The derivative is $2x^2$
- ☐ The derivative is $3x^2$
- ☐ The derivative is $\frac{1}{3}x^3$
- ☐ The derivative is $\frac{1}{2}x^2$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	21	4
1	8	41

Give the recall (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

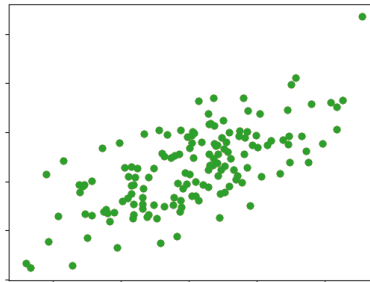


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = -0.247` and `- reg.coef_ = 1.700`. For a new observation $x = -1.335$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

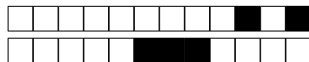
		0	1	2	3	4	5	6	7	8	9
		.									
		0	1	2	3	4	5	6	7	8	9
+		0	1	2	3	4	5	6	7	8	9
-		0	1	2	3	4	5	6	7	8	9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = -0.3$ and $\beta = 1.115$

- ☐ For $x = -1.506$, the prediction is 1.567
- ☐ The slope is positive
- ☐ For $x = 1.59$, the prediction is 1.473



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April 2026
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ENCODE 1.234

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.	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
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Instructions: No documents allowed. The use of electronic calculators and cell phones is forbidden.

Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

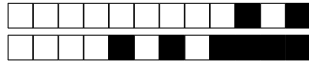
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.

Question 2 This is a numeric question where you must calculate the sum and difference of -0.4873 and -1.7111

Sum:

☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9		
.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
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☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



			0		1		2		3		4		5		6		7		8		9	
		.																				
				0		1		2		3		4		5		6		7		8		9
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	-			0		1		2		3		4		5		6		7		8		9

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

			0		1		2		3		4		5		6		7		8		9	
		.																				
				0		1		2		3		4		5		6		7		8		9
	+			0		1		2		3		4		5		6		7		8		9
	-			0		1		2		3		4		5		6		7		8		9

Standard deviation

			0		1		2		3		4		5		6		7		8		9	
		.																				
				0		1		2		3		4		5		6		7		8		9
	+			0		1		2		3		4		5		6		7		8		9
	-			0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=9$, that is x^9 ,

- ☐ The derivative is $9x^8$
- ☐ The derivative is $\frac{1}{8}x^8$
- ☐ The derivative is $\frac{1}{9}x^9$
- ☐ The derivative is $8x^8$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	36	1
1	5	9

Give the recall (tolerance ± 0.01)

			0		1		2		3		4		5		6		7		8		9	
		.																				
				0		1		2		3		4		5		6		7		8		9
	+			0		1		2		3		4		5		6		7		8		9
	-			0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

			0		1		2		3		4		5		6		7		8		9	
		.																				
				0		1		2		3		4		5		6		7		8		9
	+			0		1		2		3		4		5		6		7		8		9
	-			0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

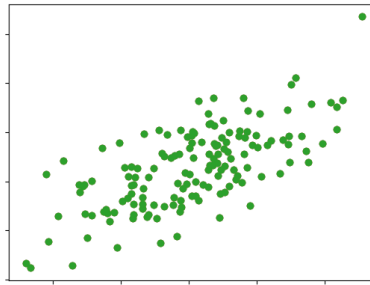


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.649` and `- reg.coef_ = 0.318`. For a new observation $x = -0.011$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

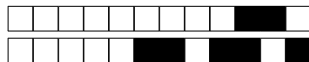
		0		1		2		3		4		5		6		7		8		9	
	.		0		1		2		3		4		5		6		7		8		9
+			0		1		2		3		4		5		6		7		8		9
-			0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = 1.665$ and $\beta = 0.896$

- ☐ For $x=2.8$, the prediction is 4.174
- ☐ For $x=-1.203$, the prediction is -1.107
- ☐ The slope is positive



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ENCODE 1.234

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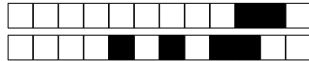
Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).

Question 2 This is a numeric question where you must calculate the sum and difference of -1.0212 and 0.8481
Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐
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Difference:



+6/2/44+

		0		1		2		3		4		5		6		7		8		9	
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		0		1		2		3		4		5		6		7		8		9	
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-		0		1		2		3		4		5		6		7		8		9	

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=5$, that is x^5 ,

- ☐ The derivative is $\frac{1}{4}x^4$
- ☐ The derivative is $\frac{1}{5}x^5$
- ☐ The derivative is $4x^4$
- ☐ The derivative is $5x^4$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	38	8
1	5	44

Give the recall (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

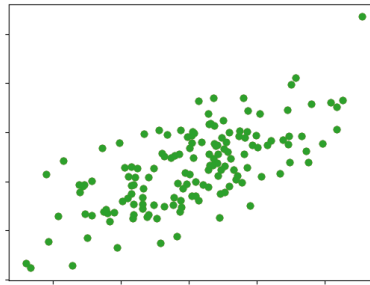


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.202` and `- reg.coef_ = 0.164`. For a new observation $x = 0.837$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

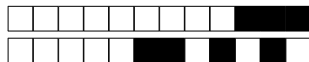
		0		1		2		3		4		5		6		7		8		9	
	.		0		1		2		3		4		5		6		7		8		9
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = -0.712$ and $\beta = -1.174$

- ☐ For $x=0.475$, the prediction is -1.512
- ☐ The slope is positive
- ☐ For $x=1.737$, the prediction is -2.751



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ENCODE 1.234

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Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

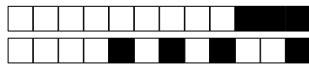
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.

Question 2 This is a numeric question where you must calculate the sum and difference of -0.1366 and 1.7033

Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
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☐ -		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



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	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

			0		1		2		3		4		5		6		7		8		9
		.																			
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Standard deviation

			0		1		2		3		4		5		6		7		8		9
		.																			
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=4$, that is x^4 ,

- ☐ The derivative is $3x^3$
- ☐ The derivative is $4x^3$
- ☐ The derivative is $\frac{1}{4}x^4$
- ☐ The derivative is $\frac{1}{3}x^3$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	17	0
1	2	37

Give the recall (tolerance ± 0.01)

			0		1		2		3		4		5		6		7		8		9
		.																			
			0		1		2		3		4		5		6		7		8		9
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

			0		1		2		3		4		5		6		7		8		9
		.																			
			0		1		2		3		4		5		6		7		8		9
	+		0		1		2		3		4		5		6		7		8		9
	-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

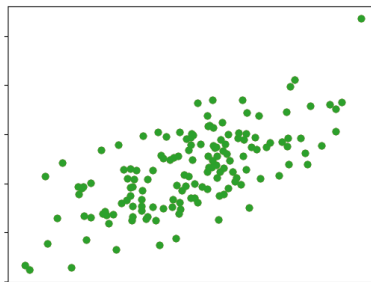


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.523` and `- reg.coef_ = 0.937`. For a new observation $x = -0.837$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9	
	.		0		1		2		3		4		5		6		7		8		9
+			0		1		2		3		4		5		6		7		8		9
-			0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = 0.098$ and $\beta = -1.571$

- ☐ For $x=0.919$, the prediction is -1.346
- ☐ The slope is positive
- ☐ For $x=-1.78$, the prediction is -1.745



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ENCODE 1.234

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☒	+	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

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Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

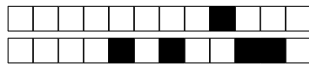
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.

Question 2 This is a numeric question where you must calculate the sum and difference of -0.1491 and 1.0056

Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	+	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



+8/2/38+

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=1$, that is x^1 ,

- ☐ The derivative is $0x^0$
- ☐ The derivative is $1x^0$
- ☐ The derivative is $\frac{1}{1}x^1$
- ☐ The derivative is $\frac{1}{0}x^0$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	42	1
1	0	39

Give the recall (tolerance ± 0.01)

+	
-	0 1 2 3 4 5 6 7 8 9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

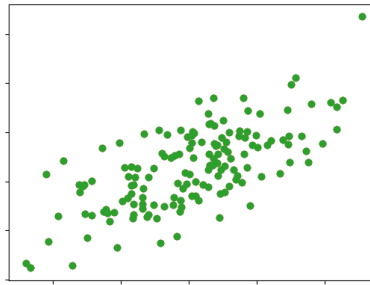


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 2.894` and `- reg.coef_ = 1.377`. For a new observation $x = 0.171$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

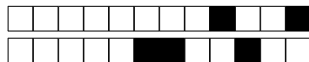
	.									
	+									
	-									

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = 0.022$ and $\beta = 1.653$

- ☐ For $x = -0.322$, the prediction is 1.646
- ☐ The slope is positive
- ☐ For $x = 1.525$, the prediction is 2.543



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CHECKED



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ENCODE 1.234

☒	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9		
☐	0	☒	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9		
.	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
☒	+	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

Instructions: No documents allowed. The use of electronic calculators and cell phones is forbidden.

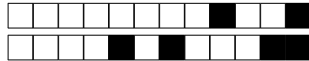
Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.
- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.

Question 2 This is a numeric question where you must calculate the sum and difference of 0.6546 and -1.3218
Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	+	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=2$, that is x^2 ,

- ☐ The derivative is $\frac{1}{2}x^2$
- ☐ The derivative is $\frac{1}{1}x^1$
- ☐ The derivative is $2x^1$
- ☐ The derivative is $1x^1$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	3	4
1	9	32

Give the recall (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

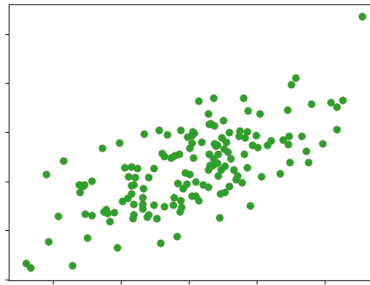


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.964` and `- reg.coef_ = 1.375`. For a new observation $x = -0.491$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

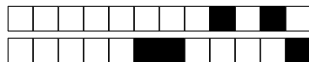
		0		1		2		3		4		5		6		7		8		9
	.																			
	+																			
	-																			

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = 2.254$ and $\beta = 0.162$

- ☐ The slope is positive
- ☐ For $x = -1.58$, the prediction is 1.998
- ☐ For $x = 0.834$, the prediction is 2.042



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CHECKED



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ENCODE 1.234

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.	☐	0	☐	1	☐	☒	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☒	+	☐	0	☐	1	☐	2	☒	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☒	4	☐	5	☐	6	☐	7	☐	8	☐	9

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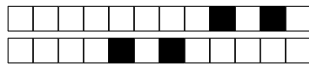
Question 1 ♣ According to the time and frequency analysis of signal x, which observations are correct?

- ☐ The analysis of the FT shows only the slow drift component without any other spectral lines.
- ☐ The Fourier Transform (FT) of the signal consists of a series of regularly spaced spectral lines, indicating a periodic underlying signal.
- ☐ The FT of the signal shows a spectral peak spacing of 0.025 in non-normalized frequencies (Hertz).
- ☐ The time signal x is approximately periodic with a period of about 40 samples.

Question 2 This is a numeric question where you must calculate the sum and difference of 1.0106 and 0.7219
Sum:

	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9	
	.	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
		☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	+	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9
☐	-	☐	0	☐	1	☐	2	☐	3	☐	4	☐	5	☐	6	☐	7	☐	8	☐	9

Difference:



+10/2/32+

		0		1		2		3		4		5		6		7		8		9	
	.																				
		0		1		2		3		4		5		6		7		8		9	
+		0		1		2		3		4		5		6		7		8		9	
-		0		1		2		3		4		5		6		7		8		9	

Question 3 Please enter the mean and standard deviation values of the time series above
Mean:

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Standard deviation

+		0		1		2		3		4		5		6		7		8		9
	.																			
-		0		1		2		3		4		5		6		7		8		9

Question 4 ♣ What is the derivative of x^n , with $n=3$, that is x^3 ,

- ☐ The derivative is $2x^2$
- ☐ The derivative is $\frac{1}{3}x^3$
- ☐ The derivative is $\frac{1}{2}x^2$
- ☐ The derivative is $3x^2$

Question 5 Consider the following *confusion matrix*:

True/Predict	0	1
0	2	2
1	8	46

Give the recall (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
	.																			
		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 6 Consider a scatter plot such as the one shown in the figure

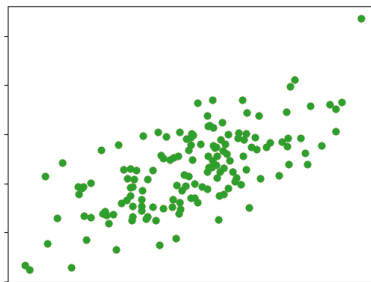
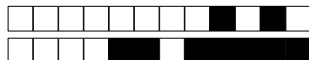


Figure: *Scatter plot*

We perform a linear regression `reg`, which yields `- reg.intercept_ = 0.505` and `- reg.coef_ = 1.001`. For a new observation $x = 0.716$, what would be the value predicted by the model? Give the accuracy (tolerance ± 0.01)

		0		1		2		3		4		5		6		7		8		9
.		0		1		2		3		4		5		6		7		8		9
+		0		1		2		3		4		5		6		7		8		9
-		0		1		2		3		4		5		6		7		8		9

Question 7 ♣ Consider the simple regression model

$$y = \alpha x + \beta$$

with $\alpha = -0.554$ and $\beta = 0.055$

- ☐ The slope is positive
- ☐ For $x = -1.699$, the prediction is 0.996
- ☐ For $x = 0.021$, the prediction is -0.553